# Advanced Labs - Theme B: Electrical Conductivity and Hall Effect of Semiconductors

Ibraheem Farzat
202118029

November 20, 2023

# Contents

**1 Experiment 1 - Day 1** **2**

1.1 Part 1: Verify Ohm's Law . . . . . 2

1.1.1 Results and Discussion . . . . . 2

1.2 Part 2: Determine carrier type . . . . . 3

**2 Experiment 2 - Day 2** **4**

**3 Experiment 3 - Day 3** **6**

# 1 Experiment 1 - Day 1

Objective: The first day's experiment aims to perform measurement using the four-terminal method and observe if Ohm's law is valid. Also to conduct the Hall effect experiment to calculate the Hall coefficient, the concentration, and mobility of free carriers. Using the results we will determine whether the carrier in the material is an electron or a hole.

## 1.1 Part 1: Verify Ohm's Law

### 1.1.1 Results and Discussion

As shown in Fig 1 we obtained values of Voltage (V) and Current (I) from the experiment. The gradient of the graph represents the resistance of the semiconductor we used in the experiment at r.t.p.

$$
\begin{aligned}
Lineequation\ y &= mx + c = 0.1268x + 0.004 \\
Resistance\ R &= Slope = 0.1268 \\
Resistivity\ \rho &= \frac{s}{l} * R = \frac{(1.5 \times 10^{-}3m)(10^{-}3m)}{1.5 \times 10^{-}3} \cdot 0.1268\Omega \approx 0.0000634\Omega \cdot m \\
Conductivity\ \sigma &= \frac{1}{\rho} \approx 15772.87\Omega^{-}1 \cdot m^{-}1
\end{aligned}
$$

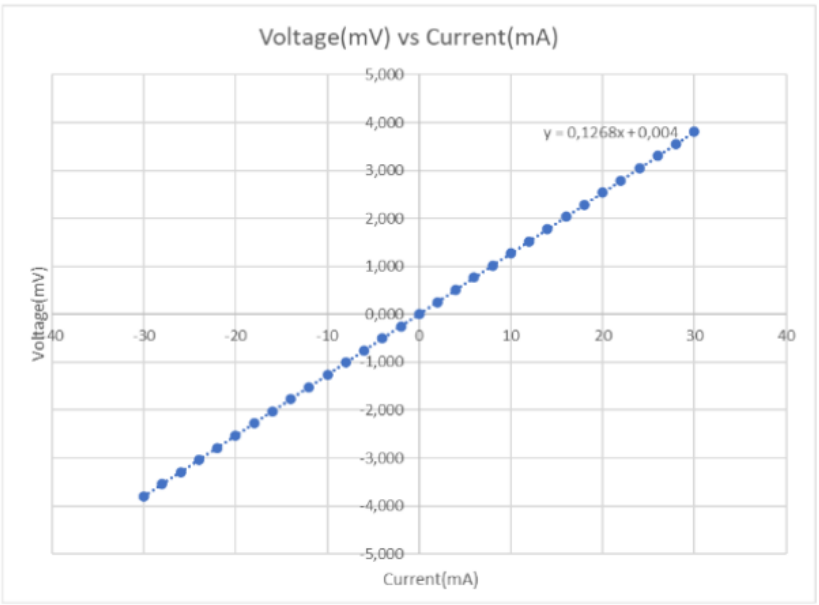



Figure 1: Voltage (mV) against Current (mA)

As we have a linear graph, we can see that Ohm's law is verified.

## 1.2 Part 2: Determine carrier type

In the second part, we varied the field strength and measured the Hall voltage and then we did the same thing but with an inverse current. We got the following figures 2 and 3.

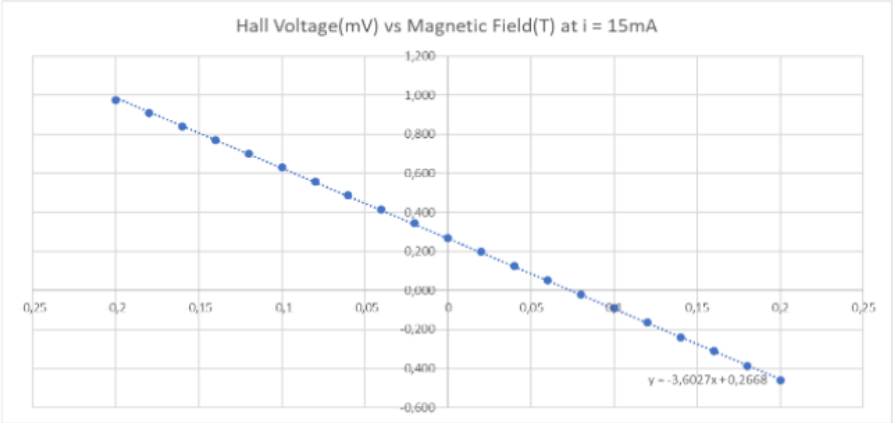



Figure 2: Hall Voltage vs Magnetic Field with positive 15mA current

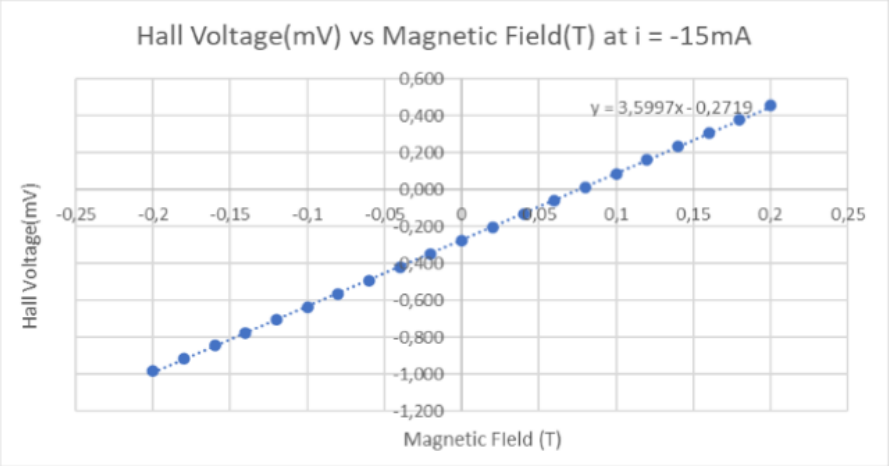



Figure 3: Hall Voltage vs Magnetic Field with negative 15mA current

We can find $R_H$ using the gradient of the Hall Voltage against the Magnetic Field strength and using the formula:

$$R_H = \frac{b \times V_y}{I_x \times B_z} = 0.00024 m^3 C^{-1}$$

Seeing that the value of $R_H$ is negative, we know that the material is an n-type semiconductor. Using the data we get:

$$\text{Carrier Mobility } \mu = \rho |R_H| = 3.78832 \frac{m^2}{V \cdot s}$$

$$\text{Carrier Relaxation Time } \tau = \frac{\mu m}{|q|} = \frac{(3.7882)(0.014)(9.11 \times 10^{-31})}{1.6 \times 10^{-19}} = 3.019 \times 10^{-13} \frac{m^3}{V \cdot s \cdot C}$$

# 2 Experiment 2 - Day 2

Objective: Measure the temperature dependency of resistivity and find the energy gap of the semiconductor. We then discuss its effect on the electric conductivity of the material.

Using the data from the experiment we find out the value of the Energy Gap $E_G$ of the semiconductor. First, we plotted resistivity against $\frac{1000}{T}$ on a logarithmic scale. We got graph 4.

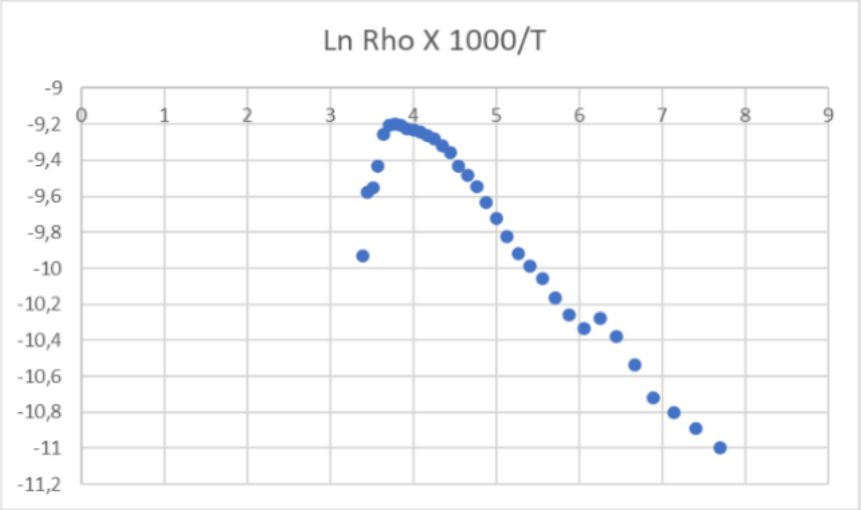



Figure 4: $\ln \rho$ against $\frac{1000}{T}$

Comparing it to the figure 5 given in the textbook we see that the region of the curve with a positive gradient is the intrinsic region of the semiconductor.

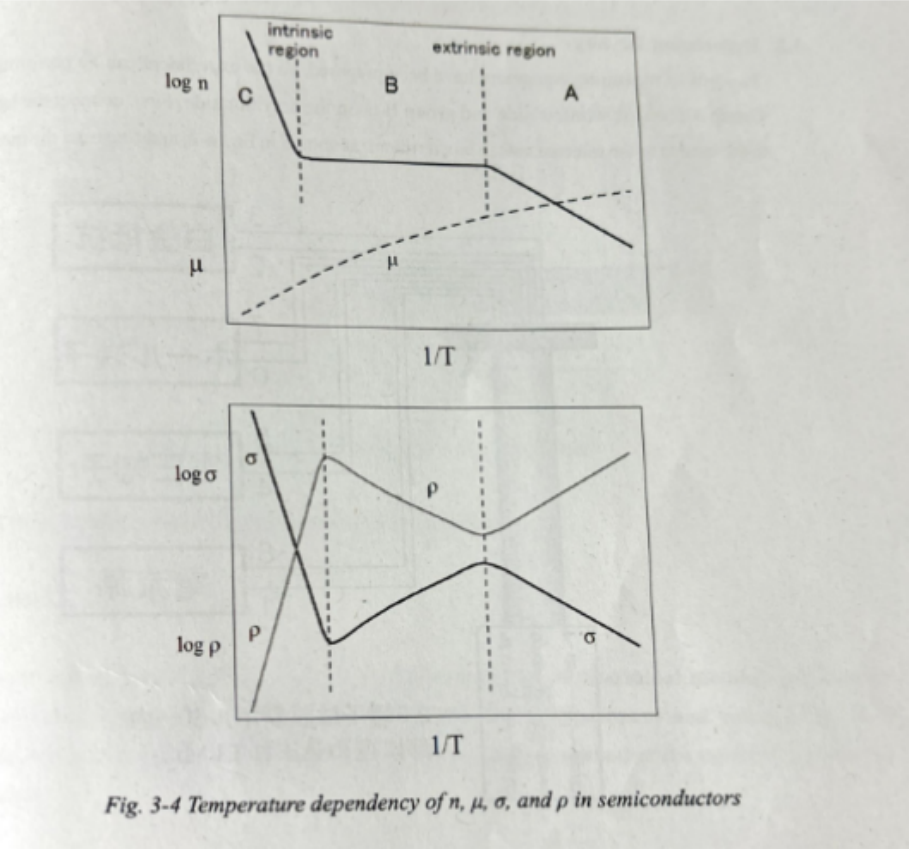



*Fig. 3-4 Temperature dependency of n, μ, σ, and ρ in semiconductors*

Figure 5: Textbook Figure

Looking at only the region with a positive gradient we get a linear graph 6. Using it we can determine the slope and find $E_G$ using the formula:

$$E_G = \frac{-slope}{2KT}$$

where $K = 8.6171 * 10^-5$ so we get $E_G = 4.955 \times 10^{-20} J = 0.15464eV$.

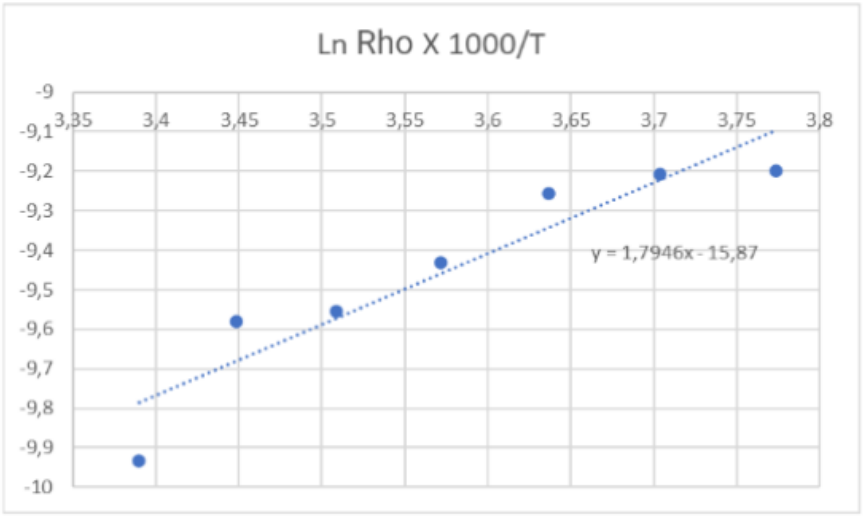



Figure 6: Truncated $\ln \rho$ against $\frac{1000}{T}$

# 3 Experiment 3 - Day 3

Objective: Investigate the temperature dependence of the Hall coefficient, determining carrier concentration and mobility's temperature dependence. Also, discover conductivity mechanisms in semiconductors and measure the Hall coefficient's response to magnetic fields at low temperatures.

From the results, we plot the Hall coefficient values against temperature, shown in graph 7.

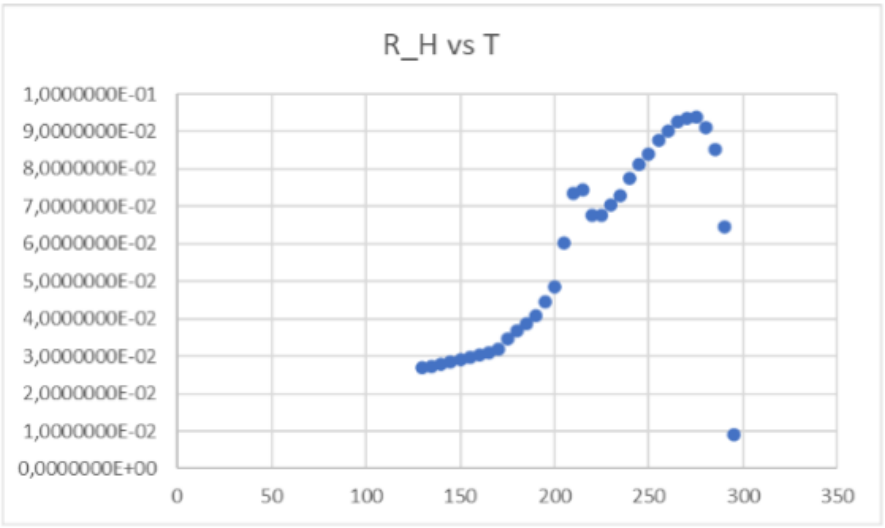



Figure 7: Hall Coefficient against Temeprature

Next we plot $R_H$ against $\frac{1000}{T}$, shown in fig 8

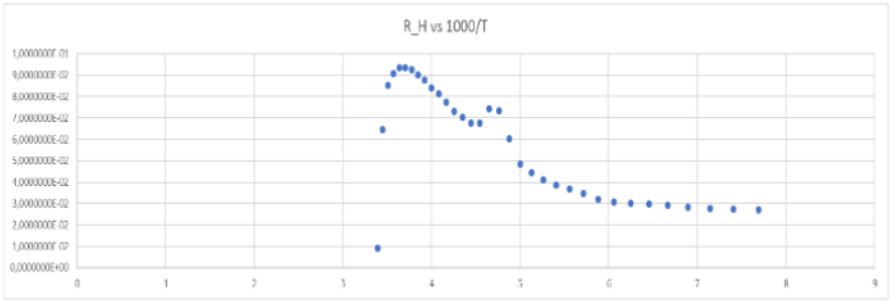



Figure 8: Hall Coefficient against inverse temperature

Then we plot the natural log of $R_H$ and n against $\frac{1000}{T}$ using a semi-logarithmic scale as shown in graphs 9 and 10.

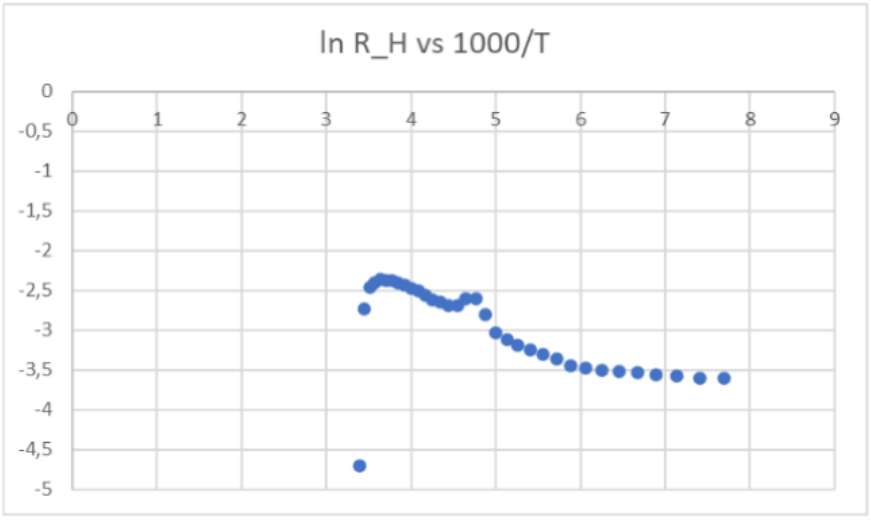



Figure 9: log natural of Hall Coefficient against 1000/T

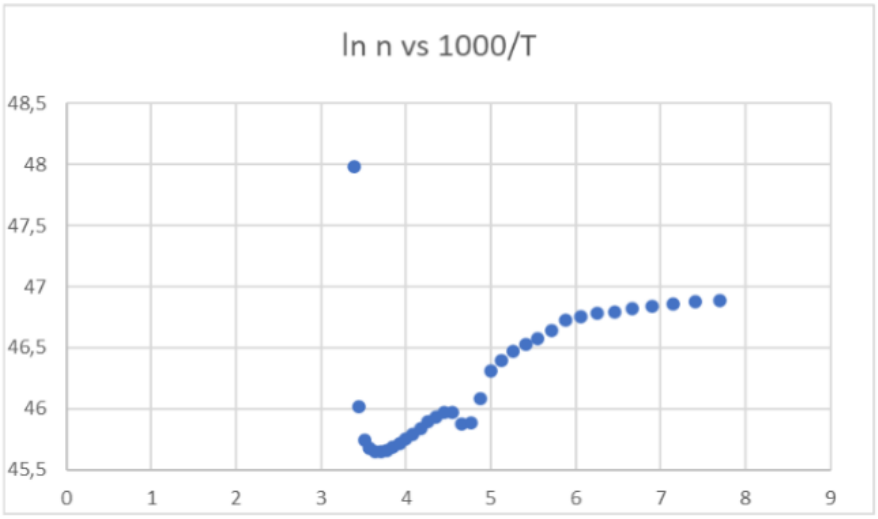



Figure 10: log natural of n against 1000/T

Finally, we find the $\ln T$ agsint $\ln u$, shown in fig 11

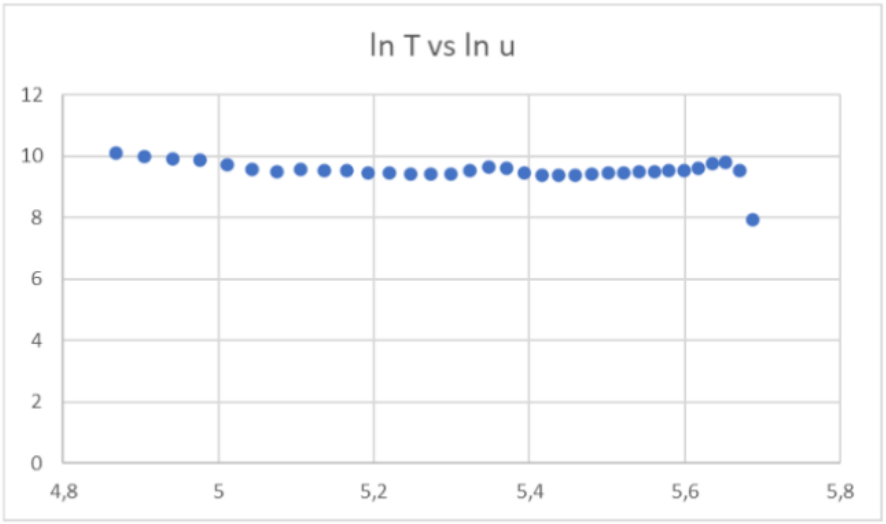



Figure 11: log natural of T against log natural of u

Analyzing the graph of $R_H$ against Temperature (T) and the natural logarithm of $R_H$ against $\frac{1}{T}$ (1000/T), similar to our Week 2 experiment, we can calculate the number of charge carriers $n$ and their mobility $\mu$ using the resistivity $\rho$ value obtained in Week 2. This helps us to identify the intrinsic region of the semiconductor. The equations for calculating carrier concentration $n$ and mobility $\mu$ are:

$$n = \frac{1}{RH \times q}$$

$$\mu = \frac{\rho}{RH}$$

Next, we refined the data from the ln RH vs 1000/T graph to find the gradient in the intrinsic

region. We then calculated the energy gap $E_G$ using the equation:

$$E_G = \frac{\text{slope}}{2kT} = 0.1110\,\text{eV}$$

The theoretical energy gap value was $E_G = 0.17\,\text{eV}$. The value we found this week ($E_G = 0.1110\,\text{eV}$) differs from last week's ($E_G = 0.15464\,\text{eV}$), likely due to experimental errors mentioned previously.

In summary, we observed that carrier mobility increases with temperature. Higher temperatures mean more carriers and higher energy, leading to more collisions and increased mobility. On the other hand, lower temperatures increase impurity scattering, reducing mobility.

For the second part, we measured the magnetic field dependency of the Hall coefficient at low temperatures, as shown in Fig 12.

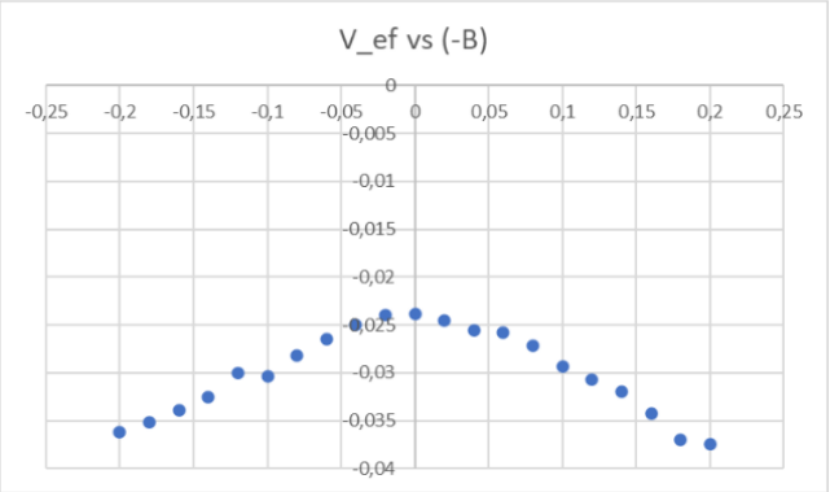



Figure 12: Graph of $V_{ef}$ against negative B

When analyzing the graph of VEF against Magnetic Field (B), we noted the Hall coefficient's dependence on the magnetic field around 100K. We calculated the Hall coefficient (RH) from the slope of this graph and used it to find the electrical conductivity $\sigma$, similar to previous experiments.

It's important to mention that the VEF vs B graph should be linear, but our data was inconsistent because it was collected while the sample was heating up after cooling down.

Finally, we used these equations to calculate mobility $\mu$ and relaxation time $\tau$:

$$\mu = \frac{RH}{\sigma}$$
$$\tau = \frac{\mu m}{q} = 2 \times 10^{-12}$$

In conclusion, the relaxation time decreases as the sample reaches extremely low temperatures.